

32) Could a set of three vectors in $\mathbb{R}^4$ span all of $\mathbb{R}^4$? Explain.
 What about n vectors in $\mathbb{R}^m$ where n is less than m ?

Solution: Let $a_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, $a_2 = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$ and $a_3 = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$ be a set of 3 vectors.

where $\forall i, 1 \leq i \leq 4, x_i, y_i, z_i \in \mathbb{R}$.
 Let $A = [a_1 \ a_2 \ a_3]$ be a 4×3 matrix.

From Theorem 4, the columns of A span all of $\mathbb{R}^4$ if and only if A has a pivot position in every row.

By defn, a pivot position of a matrix corresponds to a leading 1 of the matrix in reduced row echelon form.

$$\begin{aligned}
 &\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{bmatrix} \xrightarrow[\frac{x_2 \times \text{Row 1}}{x_1}]{\text{Row 2} = \text{Row 2} - \frac{x_2}{x_1} \text{Row 1}} \begin{bmatrix} x_1 & y_1 & z_1 \\ 0 & y_2 - \frac{x_2 y_1}{x_1} & z_2 - \frac{x_2 z_1}{x_1} \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{bmatrix} \xrightarrow[\frac{x_3 \times \text{Row 1}}{x_1}]{\text{Row 3} = \text{Row 3} - \frac{x_3}{x_1} \text{Row 1}} \\
 &\begin{bmatrix} x_1 & y_1 & z_1 \\ 0 & y_2 - \frac{x_2 y_1}{x_1} & z_2 - \frac{x_2 z_1}{x_1} \\ 0 & y_3 - \frac{x_3 y_1}{x_1} & z_3 - \frac{x_3 z_1}{x_1} \\ x_4 & y_4 & z_4 \end{bmatrix} \xrightarrow[\frac{x_4 \times \text{Row 1}}{x_1}]{\text{Row 4} = \text{Row 4} - \frac{x_4}{x_1} \text{Row 1}} \begin{bmatrix} x_1 & y_1 & z_1 \\ 0 & y_2 - \frac{x_2 y_1}{x_1} & z_2 - \frac{x_2 z_1}{x_1} \\ 0 & y_3 - \frac{x_3 y_1}{x_1} & z_3 - \frac{x_3 z_1}{x_1} \\ 0 & y_4 - \frac{x_4 y_1}{x_1} & z_4 - \frac{x_4 z_1}{x_1} \end{bmatrix} \\
 &\xrightarrow[\frac{1}{x_1}]{\text{Row 1} = \text{Row 1} \times \frac{1}{x_1}} \begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & y_2 - \frac{x_2 y_1}{x_1} & z_2 - \frac{x_2 z_1}{x_1} \\ 0 & y_3 - \frac{x_3 y_1}{x_1} & z_3 - \frac{x_3 z_1}{x_1} \\ 0 & y_4 - \frac{x_4 y_1}{x_1} & z_4 - \frac{x_4 z_1}{x_1} \end{bmatrix} \xrightarrow[\times \frac{x_1}{y_2 - \frac{x_2 y_1}{x_1}}]{\text{Row 2} = \text{Row 2} - \frac{y_2 - \frac{x_2 y_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \text{Row 2}} \\
 &\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{z_2 - \frac{x_2 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \\ 0 & 1 & \frac{z_3 - \frac{x_3 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \\ 0 & 1 & \frac{z_4 - \frac{x_4 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \end{bmatrix} \xrightarrow[\times \frac{x_1}{y_4 - \frac{x_4 y_1}{x_1}}]{\text{Row 4} = \text{Row 4} - \frac{y_4 - \frac{x_4 y_1}{x_1}}{y_4 - \frac{x_4 y_1}{x_1}} \text{Row 4}} \\
 &\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{z_2 - \frac{x_2 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \\ 0 & 1 & \frac{z_3 - \frac{x_3 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \\ 0 & 1 & \frac{z_4 - \frac{x_4 z_1}{x_1}}{y_2 - \frac{x_2 y_1}{x_1}} \end{bmatrix}
 \end{aligned}$$